

LANCASTER, PA, USA

WMO: 725116

Lat: 40.120N

Lon: 76.294W

Elev: 403

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.4	(c) 13.4	(d) -4.3	(e) 4.4	(f) 11.4	(g) 0.1	(h) 5.6	(i) 16.1	(j) 28.2	(k) 31.3	(l) 26.0	(m) 28.5	(n) 8.3	(o) 290	(p) 0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	91.1	75.1	88.5	73.5	86.0	72.3	77.4	87.2	75.8	84.4	74.6	82.3	9.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 74.6	(b) 131.5	(c) 81.7	(d) 73.1	(e) 125.0	(f) 79.9	(g) 72.2	(h) 120.9	(i) 79.0	(j) 41.0	(k) 86.8	(l) 39.5	(m) 84.3	(n) 38.2	(o) 82.5	(p) 84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.5	(b) 20.5	(c) 17.9	(e) 3.4	(f) 95.0	(g) 4.3	(h) 3.4	(i) 0.3	(j) 97.4	(k) -2.2	(l) 99.4	(m) -4.6	(n) 101.3	(o) -7.7	(p) 103.7		
			DB													
			WB													
				3.4	95.0	4.3	3.4	0.3	97.4	-2.2	99.4	-4.6	101.3	-7.7	103.7	
				2.2	79.8	4.0	1.8	-0.7	81.0	-3.0	82.1	-5.3	83.0	-8.2	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	30.0	32.4	41.0	52.1	62.3	71.1	75.1	73.5	66.9	55.1	44.4	35.3
	DBStd	17.63	9.99	9.34	9.16	8.36	7.61	5.93	4.92	4.95	6.61	8.17	8.10	8.62
	HDD50	2204	621	498	305	70	3	0	0	0	0	39	206	462
	HDD65	5224	1084	914	744	397	147	16	1	3	56	324	618	920
	CDD50	3434	2	4	26	132	383	633	779	728	506	197	38	6
	CDD65	979	0	0	0	9	62	199	314	266	112	17	0	0
	CDH74	8370	0	2	7	130	592	1710	2853	2135	825	115	1	0
	CDH80	2716	0	0	0	27	164	539	1054	686	226	20	0	0
	WSAvg	7.1	8.3	8.5	8.9	8.5	6.9	6.5	5.6	5.1	5.6	6.6	7.3	7.8
	PrecAvg	43.9	3.1	2.6	3.8	3.6	3.5	4.1	4.2	4.1	4.3	3.8	3.1	3.6
(15) Precipitation	PrecMax	62.0	6.5	5.7	6.5	7.8	5.9	11.1	8.4	7.8	13.3	8.9	5.7	6.3
	PrecMin	29.6	1.1	0.5	1.0	1.2	1.0	1.2	1.2	1.5	0.4	0.4	0.9	0.6
	PrecStd	7.1	1.4	1.2	1.4	1.6	1.3	2.0	1.8	1.7	3.0	2.2	1.3	1.3
	PrecStd	7.1	1.4	1.2	1.4	1.6	1.3	2.0	1.8	1.7	3.0	2.2	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.7	73.8	83.8	88.8	92.0	96.1	93.2	90.4	82.4	70.2	64.0
		MCWB	56.7	56.8	59.6	66.0	71.4	74.5	76.5	75.7	74.0	70.1	60.6	58.8
	2%	DB	55.0	56.9	66.1	78.2	84.4	89.0	91.7	89.9	85.9	77.1	65.7	57.5
		MCWB	50.3	49.6	55.3	62.5	69.7	73.2	75.9	74.8	71.4	67.5	58.7	53.7
	5%	DB	48.4	50.3	61.1	72.3	81.0	85.9	88.8	86.6	82.0	72.7	61.9	52.4
		MCWB	44.8	43.5	51.9	59.9	67.6	71.9	74.3	73.2	69.3	64.7	56.0	49.3
	10%	DB	44.1	45.2	55.8	67.3	76.6	82.7	85.9	83.8	79.0	69.2	57.4	47.8
		MCWB	40.1	40.3	48.1	56.8	65.6	70.0	72.9	71.9	67.8	61.6	52.2	43.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	59.1	61.9	69.1	74.2	76.9	80.7	78.9	77.0	73.2	64.0	60.4
		MCDB	64.0	63.6	70.9	77.3	84.8	87.9	91.2	89.3	86.2	77.2	67.8	62.8
	2%	WB	50.8	50.6	57.6	65.2	71.7	75.2	77.9	77.2	74.3	69.7	60.6	54.7
		MCDB	53.7	55.9	63.5	74.3	81.3	84.8	87.9	86.4	80.4	75.5	64.1	57.3
	5%	WB	45.5	44.5	53.6	61.9	69.4	73.7	76.1	75.4	72.6	66.4	57.0	49.8
		MCDB	48.2	48.6	59.3	70.2	77.9	82.4	85.1	82.7	78.4	71.4	60.8	52.2
	10%	WB	40.5	40.6	49.1	58.4	67.3	72.2	74.8	73.9	70.8	62.7	53.6	44.5
		MCDB	43.6	45.2	55.3	65.9	74.6	80.3	82.9	80.6	76.4	67.1	56.9	47.2
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	15.8	18.1	20.3	19.7	18.8	18.6	18.3	19.0	18.4	17.1	14.4
		MCDBR	20.6	23.2	26.3	27.5	24.9	22.0	22.0	21.2	22.8	22.8	21.9	19.4
		MCWBR	17.3	17.3	17.4	15.8	12.2	9.4	8.6	8.7	10.3	12.7	15.4	16.8
		MCDBR	19.3	20.9	24.0	24.3	22.0	19.4	18.6	17.8	17.6	18.9	17.8	18.3
	5% WB	MCWBR	17.3	17.3	17.9	15.5	11.8	9.5	8.4	8.4	9.4	12.1	14.5	17.3
		MCWBR	17.3	17.3	17.9	15.5	11.8	9.5	8.4	8.4	9.4	12.1	14.5	17.3
(40) Clear-Sky Solar Irradiance	taub		0.309	0.326	0.358	0.392	0.431	0.497	0.499	0.499	0.415	0.365	0.326	0.307
	taud		2.435	2.363	2.356	2.286	2.211	2.066	2.072	2.059	2.264	2.414	2.487	2.484
	Ebn at Noon		271	281	283	279	270	252	249	245	261	264	263	263
	Edh at Noon		25	31	36	41	45	52	51	51	39	30	24	23
(44) All-Sky Solar Radiation	RadAvg		591	873	1163	1506	1687	1863	1885	1645	1349	961	679	505
	RadStd		50	107	88	138	202	153	96	95	171	106	62	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	N/A
(47) Regional (72 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+221	+124

Nomenclature: See separate page